ML-driven PVT-variation-aware leakage estimation - DVD course project

Summary - final models (40,000 samples per case). Gate level (18 gate/input cases x 200 PVT corners): mean accuracy **99.19 %**, median error 0.53 %, 95th percentile 2.55 %, bias +0.28 %. 8-bit parity checker (256 vectors x 200 corners): mean accuracy **99.38 %**, median error 0.45 %, 95th percentile 1.96 %. Transistor models (test sets, pooled median relative error): nmos_on 0.07 %, nmos_off 0.08 %, pmos_on 0.07 %, pmos_off 0.39 %.

With the specified 10,000 samples per case (deliverable datasets, `data/`): gate level mean accuracy **98.36 %**, median error 1.09 %, 95th percentile 5.24 %; 8-bit parity checker (256 vectors x 200 corners): mean accuracy **98.59 %**, median error 1.09 %, 95th percentile 4.03 %. Transistor models (test sets, pooled over Id, Ig, Is, Ib, median relative error): nmos_on 0.24 %, nmos_off 0.24 %, pmos_on 0.19 %, pmos_off 1.06 %.

Sections 2-5 walk through the pipeline on the specified 10,000-sample datasets. Section 6 gives the final models: the same pipeline (same sampling, features, model selection and the same 200 gate corners and circuit SPICE reference) with 40,000 samples per case. Section 7 shows why: the remaining error is a data (variance) limit, and accuracy grows with the dataset size.

1. Setup

- Assigned model file: **32nm_MGK** (PTM 32 nm metal-gate / high-k, BSIM4); nominal VDD **0.9 V** (VDD drawn in 0.81-0.99 V); L = Lmin = 32 nm, W = 1-8 Lmin; test circuit: **8-bit parity checker**.
- Simulator: ngspice-36 : Circuit level simulation program The U. C. Berkeley CAD Group, run on the Ada cluster (Ubuntu 22.04). ML: PyTorch (MLP) and scikit-learn (other model types); fixed seed 2024.
- Conventions: terminal currents are currents INTO the terminal (Id, Ig, Is, Ib = -I(V_x) of the source on that terminal). Gate leakage = current delivered by every ideal source held at VDD (supply and inputs at logic 1) = static power / VDD; the same definition is used for SPICE and ML.
- Transistor netlists: one voltage source per terminal; `.options gmin=1e-18 abstol=1e-18 reltol=1e-6 vntol=1e-9`; the five process parameters (vth0, toxe, ndep, u0, xj) are passed with `.param` into a parametrised copy of the model file, verified identical to the original at nominal (`results/00_setup.txt`). AD = AS = W*2Lmin, PD = PS = 2(W + 2Lmin).
- Note on toxe: in the model file toxe = toxp + dtox exactly (NMOS 0.75 = 0.50 + 0.25 nm, PMOS 0.77 = 0.50 + 0.27 nm). Varying toxe by +-10 % as specified, with toxp and dtox kept from the file, breaks that identity, so ngspice reports "toxe != toxp + dtox; dtox ignored" and keeps toxp from the file: the toxe variation acts on the toxe-based terms (e.g. gate tunnelling, Coxe) only. The SPICE references and the training data use the same model, so this does not bias the comparison.
- Gates (static CMOS, width unit WU = 2 Lmin): INV N = 1 WU, P = 2 WU; NAND2 N = 2 WU (series), P = 2 WU; NOR2 N = 1 WU, P = 4 WU (series); AND2 = NAND2 + INV, OR2 = NOR2 + INV with the same sizes. Internal series nodes: x (NAND2), y (NOR2).
- Parity checker: XOR(a, b) = AND2(OR2(a, b), NAND2(a, b)); a balanced tree of 7 XORs = 21 gates, 112 transistors, one netlist.

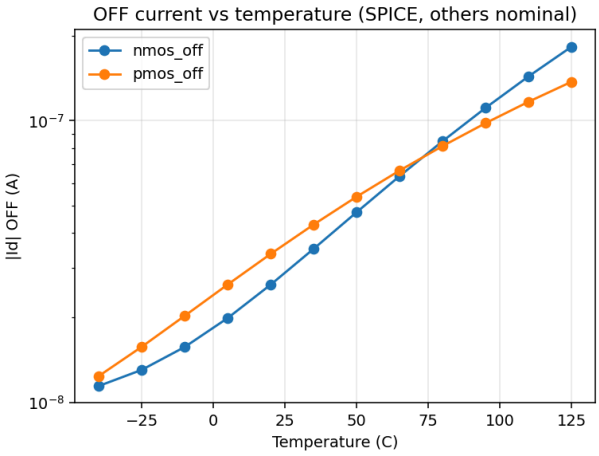
2. Data generation

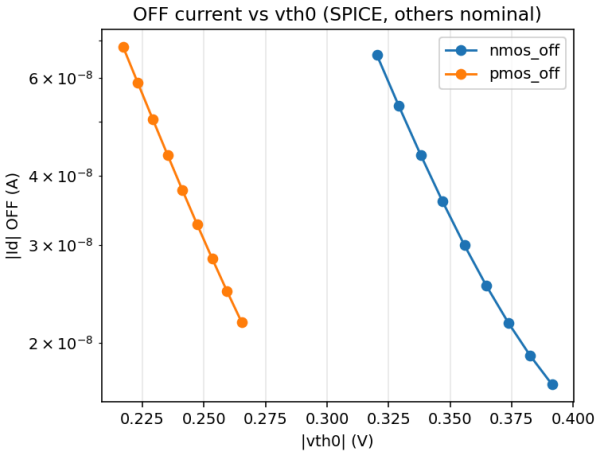
- 10000 samples per case (nmos_on: 10000, nmos_off: 10000, pmos_on: 10000, pmos_off: 10000). Every parameter is drawn independently and uniformly: VDD first (nominal +-10 %), then Vd, Vs in 0..VDD, T in -40..125 C, W in 1-8 Lmin, vth0, toxe, ndep, u0, xj +-10 % around the file's nominal values (separate NMOS and PMOS draws). Vg and Vb are fixed per case: NMOS ON Vg = VDD, NMOS OFF Vg = 0, PMOS ON Vg = 0, PMOS OFF Vg = VDD; NMOS body at 0, PMOS body at VDD. Non-converged samples are redrawn.
- Split: 70 / 15 / 15 (train / validation / test), fixed seed (random_state = 2024).
- The gate and parity-checker SPICE results (Parts B and C) are used for evaluation only: no training or model-selection script reads them (checked in `results/10_requirements_check.txt`). Model types are chosen on the device validation sets.
- The final models (section 6) use the same procedure with 40,000 samples per case (`data_40k/`); the dataset-size study (section 7) repeats it at 2,500 / 5,000 / 20,000 samples per case.

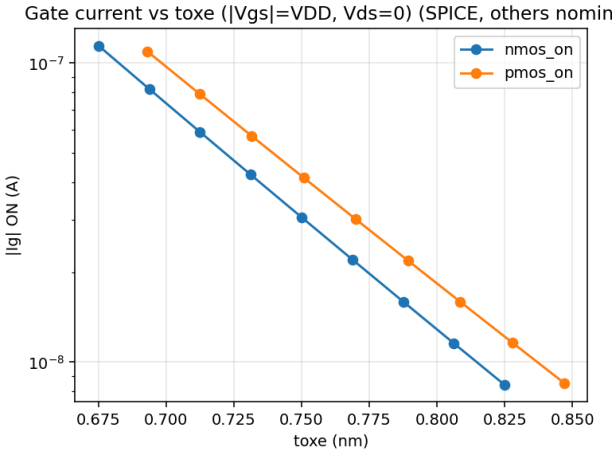
Data check (Kirchhoff residual and physical trends)

```
Kirchhoff check: KCL = id+ig+is+ib ; rel = |KCL| / (|id|+|ig|+|is|+|ib|)
nmos_on: n=10000  max|KCL|=1.375e-15 A  median|KCL|=7.198e-17 A  max rel=4.031e-05  median rel=6.868e-13  frac(rel<1e-3)=1.0000
nmos_off: n=10000  max|KCL|=5.438e-22 A  median|KCL|=3.423e-24 A  max rel=4.414e-15  median rel=3.297e-16  frac(rel<1e-3)=1.0000
pmos_on: n=10000  max|KCL|=1.057e-16 A  median|KCL|=7.772e-18 A  max rel=1.460e-06  median rel=1.675e-13  frac(rel<1e-3)=1.0000
pmos_off: n=10000  max|KCL|=7.622e-15 A  median|KCL|=3.105e-16 A  max rel=5.845e-05  median rel=2.847e-08  frac(rel<1e-3)=1.0000
nmos_off: Id(T) increases monotonic: True; Id vs |vth0| decreases monotonic: True
pmos_off: Id(T) increases monotonic: True; Id vs |vth0| decreases monotonic: True
nmos_on: |Ig| vs toxe decreases monotonic: True
pmos_on: |Ig| vs toxe decreases monotonic: True
```

Controlled sweeps (one parameter varied, all others nominal):







3. Models

- Inputs: the 12 sample parameters, turned into physics features (same for every model type): thermal voltage vT, terminal-voltage differences (PMOS voltages mirrored to NMOS polarity), the same divided by vT, gate overdrive (Vg* - |vth0|)/vT, the subthreshold drain-bias factor 1 - exp(-|Vds|/vT), oxide-field terms Vgs/toxe and Vgd/toxe, ln W, ln toxe, ln ndep, ln u0, xj.
- Why the bounded drain-bias factor: inside gates the ON transistors sit at |Vds| ~ 0, beyond the 0.1 % quantile of the uniform training data, where ln(1 - exp(-|Vds|/vT)) runs far outside its training range. On an independent rail-bias validation set (Vd, Vs snapped to 0 / VDD, own seed, single-FET SPICE) the median Ig error at Vds = 0 was 41.2-133.8 % with the logarithm and 3.7-10.1 % with the bounded form, while the ordinary validation median error changed by at most 0.03 percentage points (`experiments/exp7_bounded_vds_feature.py`). Re-weighting the training loss toward the rails instead gave a smaller and less stable gain (`experiments/exp5_rail_weighting.py`).
- Target per terminal: y = asinh(I / I0_t), I0_t = 1 % of the training median |I_t|. This is logarithmic (so the loss tracks the relative error) for |I| >> I0_t and stays linear and smooth where a current changes sign (Vds near 0).
- Four model types: polynomial ridge regression (degree 2 and 3, RidgeCV), random forest (500 trees), histogram gradient boosting (one per terminal, early stopping), and an MLP (4 hidden layers x 128, SiLU, Adam with cosine learning rate for 2000 epochs, then L-BFGS, ensemble of 5 seeds, float64).
- Selection: per model type the best hyper-parameter variant on validation; per case the model type with the smallest validation (median + 95th percentile) of the pooled relative error. Relative error = |dI| / max(|I_SPICE|, 1 fA).

Model comparison (validation set, pooled over Id, Ig, Is, Ib)

model	nmos_on R2	nmos_on median %	nmos_on p95 %	nmos_off R2	nmos_off median %	nmos_off p95 %	pmos_on R2	pmos_on median %	pmos_on p95 %	pmos_off R2	pmos_off median %	pmos_off p95 %
poly_ridge	0.991	16.48	252.9	0.9431	11.15	129.3	0.9917	13.99	189.1	0.8921	12.08	116.7
random_forest	0.9883	19.44	179	0.9653	15.35	79.35	0.99	17.59	147.1	0.9169	14.72	70.86
grad_boost	0.998	6.708	51.62	0.99	4.377	37.15	0.9983	6.025	42.83	0.9319	4.285	29.47
mlp	0.9996	0.232	3.78	0.9938	0.232	2.78	0.9999	0.182	3.04	0.9264	1.045	8.13

Chosen models (test set)

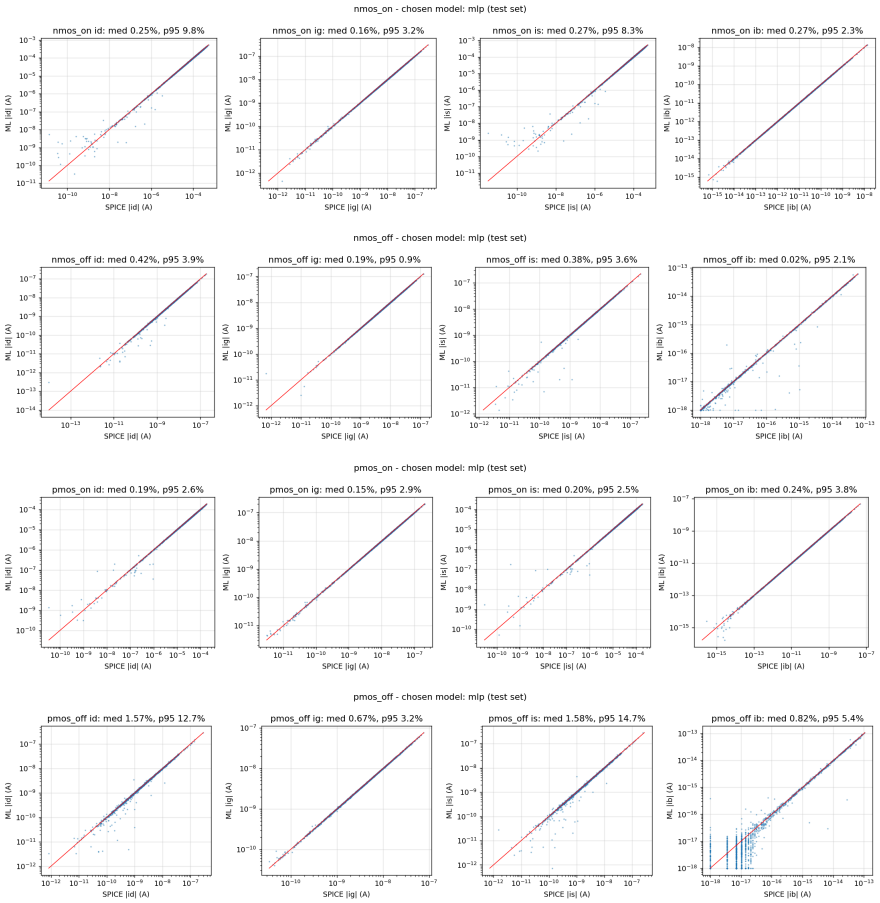
Selection rule: smallest validation (median + p95) pooled error. Per case:

- nmos_on: **mlp**, validation median 0.232 % / p95 3.78 % against the runner-up grad_boost (6.708 % / 51.62 %).
- nmos_off: **mlp**, validation median 0.232 % / p95 2.78 % against the runner-up grad_boost (4.377 % / 37.15 %).
- pmos_on: **mlp**, validation median 0.182 % / p95 3.04 % against the runner-up grad_boost (6.025 % / 42.83 %).
- pmos_off: **mlp**, validation median 1.045 % / p95 8.13 % against the runner-up grad_boost (4.285 % / 29.47 %).

Median / 95th-percentile relative error (%) per terminal; `pooled_|Vds|>5%VDD` excludes the near-zero-Vds samples where the channel current itself crosses zero:

case	chosen model	id median %	id p95 %	ig median %	ig p95 %	is median %	is p95 %	ib median %	ib p95 %	pooled median %	pooled p95 %	pooled_|Vds|>5%VDD median %	pooled_|Vds|>5%VDD p95 %	predicted KCL residual (median)
nmos_on	mlp	0.253	9.79	0.155	3.19	0.266	8.35	0.27	2.27	0.236	3.97	0.213	1.6	0.0003734
nmos_off	mlp	0.419	3.85	0.185	0.87	0.382	3.57	0.019	2.06	0.238	2.57	0.234	2.26	0.001133
pmos_on	mlp	0.193	2.6	0.147	2.93	0.204	2.52	0.238	3.78	0.192	3.11	0.173	1.59	0.0003367
pmos_off	mlp	1.571	12.69	0.673	3.15	1.577	14.72	0.818	5.37	1.063	8.74	1.043	7.79	0.004653

Predicted-vs-SPICE plots (test set, log axes):



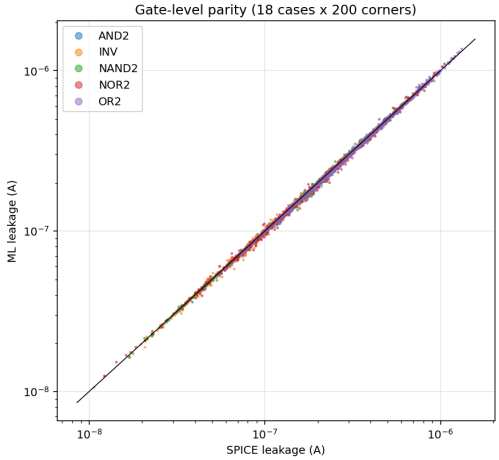
4. Gate results (18 cases x 200 corners)

SPICE simulates each complete gate; the ML estimate takes every transistor's terminal voltages from the SPICE-recorded output and internal nodes, picks the ON or OFF model by the gate voltage, predicts the four terminal currents and forms the leakage as sum over devices of [(Vd-Vs) Id + (Vg-Vs) Ig + (Vb-Vs) Ib] / VDD. By Tellegen's theorem this equals the current drawn from the VDD-level sources exactly (checked with 0 V ammeters on every terminal: relative difference <= 2.8e-08, `experiments/exp1_power_formula.py`); it avoids taking small differences of large ON-device currents.

Metrics in every gate and circuit table, per row over its n simulations: relative error e = (ML - SPICE) / SPICE; bias = median e; median error = median |e|; p95 error = 95th percentile of |e|; **mean accuracy = 100 % - mean |e|**.

gate	pattern	n	spice_median	ml_median	bias_pct	median_err_pct	p95_err_pct	mean_accuracy_pct
INV	0	200	8.104e-08	8.094e-08	-0.1841	0.924	3.013	98.86
INV	1	200	9.759e-08	9.628e-08	-0.3606	1.903	8.212	97.28
NAND2	00	200	1.117e-07	1.127e-07	-0.01246	0.7019	2.456	99.05
NAND2	01	200	1.952e-07	1.943e-07	-0.1537	0.7731	2.81	98.9
NAND2	10	200	8.999e-08	8.913e-08	0.1352	0.7491	2.813	99.03
NAND2	11	200	2.647e-07	2.638e-07	-0.3261	1.48	7.014	97.79
NOR2	00	200	2.667e-07	2.662e-07	0.01088	0.9886	3.191	98.8
NOR2	01	200	2.716e-07	2.71e-07	-0.9104	1.783	7.893	97.34
NOR2	10	200	1.288e-07	1.25e-07	-1.345	2.34	7.961	97.08
NOR2	11	200	9.102e-08	9.105e-08	0.23	1.152	5.016	98.29
AND2	00	200	2.245e-07	2.253e-07	-0.3567	0.8597	3.886	98.68
AND2	01	200	2.938e-07	2.942e-07	-0.2384	1.008	4.073	98.65
AND2	10	200	1.967e-07	1.926e-07	-0.3345	1.117	3.873	98.57
AND2	11	200	3.583e-07	3.589e-07	-0.33	1.109	4.778	98.33
OR2	00	200	3.735e-07	3.728e-07	-0.3804	0.8844	3.333	98.81
OR2	01	200	3.551e-07	3.534e-07	-0.7573	1.495	6.469	97.86
OR2	10	200	2.15e-07	2.105e-07	-1.006	1.516	5.093	98.09
OR2	11	200	1.835e-07	1.828e-07	-0.1519	0.7563	2.762	99

Overall: mean accuracy **98.36 %**, median error 1.09 %, 95th percentile 5.24 %, bias -0.31 % (n = 3600). Most accurate case: NAND2 00 (median error 0.702 %); least accurate: NOR2 10 (2.34 %).



Extra (not required): ML-only node voltages. Instead of reading x, y and the INV input of AND2/OR2 from SPICE, the internal node voltages are solved from the ML currents by KCL (grid search plus bisection, Gauss-Seidel over coupled nodes). With these self-consistent voltages the gate leakage mean accuracy is 97.80 % (median error 1.40 %). Node-voltage errors vs SPICE:

gate	pattern	node	spice_median_V	ml_median_V	abs_err_median_mV	abs_err_p95_mV	bracketed_frac
NAND2	0	x	0.07888	0.07886	0.6601	2.822	1
NAND2	1	x	0.0001109	0.0006431	3.079	7.052	1
NAND2	10	x	0.731	0.7319	1.487	4.697	1
NAND2	11	x	0.0003213	0.0003876	0.08945	1.46	1
NOR2	0	y	0.8945	0.8944	0.3301	1.034	1
NOR2	1	y	0.8947	0.8928	2.335	4.749	1
NOR2	10	y	0.1362	0.1343	1.998	4.872	1
NOR2	11	y	0.8244	0.8244	2.236	6.501	1
AND2	0	x	0.07888	0.07896	0.7197	3.122	1
AND2	0	n	0.8953	0.8957	1.338	5.552	1
AND2	1	x	0.0001109	0.0006407	3.079	7.054	1
AND2	1	n	0.8942	0.8947	1.1	4.679	1
AND2	10	x	0.731	0.732	1.545	4.65	1
AND2	10	n	0.8945	0.895	1.28	5.1	1
AND2	11	x	0.0004164	0.001035	2.016	5.2	1
AND2	11	n	0.0007149	0.002183	5.406	12.01	1
OR2	0	y	0.8944	0.8935	1.671	3.527	1
OR2	0	n	0.8937	0.8914	4.17	8.334	1
OR2	1	y	0.8947	0.8928	2.33	4.749	1
OR2	1	n	0.00063	0.001938	3.236	7.898	1
OR2	10	y	0.1363	0.1343	1.67	4.703	1
OR2	10	n	0.0005145	0.001759	3.201	7.8	1
OR2	11	y	0.8244	0.8242	2.144	6.529	1
OR2	11	n	0.0001816	0.0009673	3.054	7.505	1

5. Test-circuit results (8-bit parity checker, 256 vectors x 200 corners)

SPICE simulates the full 112-transistor netlist for every input vector at the same 200 corners (output logic checked in every run). The hierarchical ML estimate evaluates the logic value at each gate's inputs, looks up that gate's ML leakage at the corner and sums the 21 gate leakages.

grp	n	spice_median	ml_median	bias_pct	median_err_pct	p95_err_pct	mean_accuracy_pct
parity8_ALL	51200	5.351e-06	5.324e-06	-0.5468	1.086	4.034	98.59
SPICE-gate-sum vs SPICE circuit	51200	5.351e-06	5.346e-06	-0.1561	0.1605	0.6546	99.77
ML with ML-solved node voltages (extra)	51200	5.351e-06	5.356e-06	-0.5224	1.263	4.278	98.4

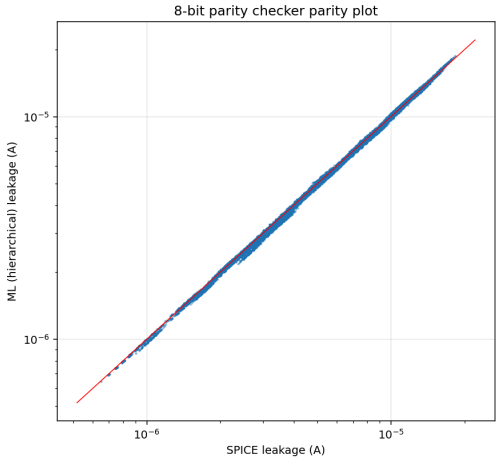
Compared with the gate level: circuit mean accuracy 98.59 % vs 98.36 % for the gates, median error 1.09 % vs 1.09 %, 95th percentile 4.03 % vs 5.24 %. Summing 21 gate estimates averages out part of the independent per-gate error, while the common (bias) part adds up.

The second row isolates the error of the hierarchical assumption itself: summing **SPICE** gate leakages (no ML) differs from the full-circuit SPICE result by only 0.16 % median (99.77 % mean accuracy), so gate-to-gate loading is small and the circuit-level error is almost entirely the gate-level ML error.

By number of ones in the input vector:

hw	n	spice_median	ml_median	bias_pct	median_err_pct	p95_err_pct	mean_accuracy_pct
0	200	5.433e-06	5.437e-06	-0.2595	0.8109	3.243	98.92
1	1600	5.501e-06	5.504e-06	-0.4805	0.9383	3.765	98.73
2	5600	5.42e-06	5.397e-06	-0.5132	0.9983	3.803	98.69
3	11200	5.45e-06	5.431e-06	-0.5935	1.076	4.021	98.58

hw	n	spice_median	ml_median	bias_pct	median_err_pct	p95_err_pct	mean_accuracy_pct
4	14000	5.309e-06	5.276e-06	-0.5085	1.068	3.973	98.61
5	11200	5.342e-06	5.301e-06	-0.6178	1.169	4.187	98.51
6	5600	5.188e-06	5.156e-06	-0.5065	1.119	4.121	98.57
7	1600	5.22e-06	5.142e-06	-0.5472	1.131	4.051	98.57
8	200	4.982e-06	5.007e-06	-0.3123	0.9146	3.506	98.78



6. Final models: 40,000 samples per case

The pipeline of sections 2-5 unchanged, run with 40,000 samples per case: datasets `data_40k/` (4 x 40,000 rows, the same Section 4 sampling and ranges, checked in `results/10_requirements_check.txt`), models `saved_models_40k/`, tables `results_40k/`, plots `plots_40k/`. The 70/15/15 split, the four model types and the selection rule are the same; the gate corners and the full-circuit SPICE reference are identical to the main run, so every number here is directly comparable with sections 4 and 5. Chosen model types: nmos_on mlp, nmos_off mlp, pmos_on mlp, pmos_off mlp. The size is chosen by the same rule as the model types, on the device validation sets: the best validation median error falls with every doubling for all four cases and is lowest at 40,000 samples per case, the largest size run (table in section 7).

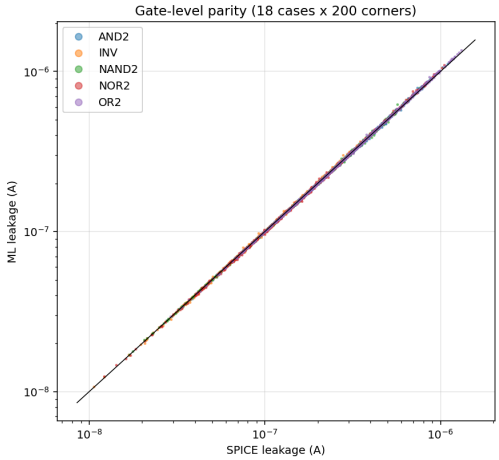
Chosen models (test set)

case	chosen model	id median %	id p95 %	ig median %	ig p95 %	is median %	is p95 %	ib median %	ib p95 %	pooled median %	pooled p95 %	pooled_|Vds|>5%VDD median %	pooled_|Vds|>5%VDD p95 %	predicted KCL residual (median)
nmos_on	mlp	0.077	1.48	0.061	0.71	0.076	1.42	0.083	0.73	0.074	0.94	0.068	0.52	5.984e-05
nmos_off	mlp	0.137	0.93	0.063	0.29	0.135	0.97	0.004	0.61	0.08	0.7	0.078	0.65	0.0004057
pmos_on	mlp	0.067	0.65	0.053	0.63	0.066	0.63	0.078	1.15	0.065	0.76	0.06	0.41	6.044e-05
pmos_off	mlp	0.435	3.11	0.169	0.86	0.428	3.27	0.688	2.56	0.386	2.43	0.378	2.28	0.001084

Gate results (18 cases x 200 corners)

gate	pattern	n	spice_median	ml_median	bias_pct	median_err_pct	p95_err_pct	mean_accuracy_pct
INV	0	200	8.104e-08	8.156e-08	0.4101	0.4484	3.094	99.16
INV	1	200	9.759e-08	9.763e-08	0.5484	0.8224	3.129	98.82
NAND2	00	200	1.117e-07	1.122e-07	0.3114	0.3225	0.9369	99.61
NAND2	01	200	1.952e-07	1.944e-07	0.1179	0.4046	2.78	99.24
NAND2	10	200	8.999e-08	9.018e-08	0.5143	0.515	1.267	99.42
NAND2	11	200	2.647e-07	2.646e-07	0.1433	0.8028	3.604	98.92
NOR2	00	200	2.667e-07	2.667e-07	0.3738	0.4625	2.597	99.16
NOR2	01	200	2.716e-07	2.708e-07	0.2159	0.5832	2.99	99.08
NOR2	10	200	1.288e-07	1.293e-07	-0.001327	0.6919	2.638	99.06
NOR2	11	200	9.102e-08	9.057e-08	-0.3483	0.6762	2.56	99.06
AND2	00	200	2.245e-07	2.248e-07	0.2887	0.4828	1.757	99.35
AND2	01	200	2.938e-07	2.931e-07	0.1707	0.5441	2.611	99.17
AND2	10	200	1.967e-07	1.962e-07	0.4168	0.5247	2.043	99.27
AND2	11	200	3.583e-07	3.58e-07	0.2378	0.6692	2.889	99.08
OR2	00	200	3.735e-07	3.75e-07	0.3331	0.4351	2.467	99.22
OR2	01	200	3.551e-07	3.572e-07	0.2257	0.5301	2.58	99.18
OR2	10	200	2.15e-07	2.146e-07	0.2317	0.4631	2.155	99.27
OR2	11	200	1.835e-07	1.824e-07	0.05977	0.437	1.881	99.36

Overall: mean accuracy **99.19 %**, median error 0.53 %, 95th percentile 2.55 %, bias +0.28 % (n = 3600); with the specified 10,000 samples per case: 98.36 %, 1.09 %, 5.24 %. Most accurate case: NAND2 00 (median error 0.323 %); least accurate: INV 1 (0.822 %). With ML-solved internal node voltages (extra, as in section 4): 99.13 % mean accuracy.

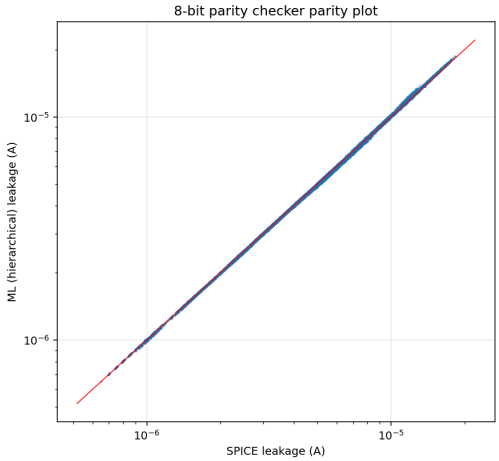


Test-circuit results (8-bit parity checker, 256 vectors x 200 corners)

grp	n	spice_median	ml_median	bias_pct	median_err_pct	p95_err_pct	mean_accuracy_pct
parity8_ALL	51200	5.351e-06	5.34e-06	0.09936	0.4468	1.961	99.38
SPICE-gate-sum vs SPICE circuit	51200	5.351e-06	5.346e-06	-0.1561	0.1605	0.6546	99.77
ML with ML-solved node voltages (extra)	51200	5.351e-06	5.344e-06	0.1084	0.4922	2.052	99.34

Mean accuracy **99.38 %** (specified 10,000 samples per case: 98.59 %). By number of ones in the input vector:

hw	n	spice_median	ml_median	bias_pct	median_err_pct	p95_err_pct	mean_accuracy_pct
0	200	5.433e-06	5.432e-06	0.2576	0.4434	1.887	99.35
1	1600	5.501e-06	5.515e-06	0.1481	0.4141	1.846	99.39
2	5600	5.42e-06	5.414e-06	0.1408	0.4326	1.907	99.39
3	11200	5.45e-06	5.425e-06	0.07955	0.4471	1.931	99.39
4	14000	5.309e-06	5.3e-06	0.1119	0.4421	1.982	99.38
5	11200	5.342e-06	5.317e-06	0.06805	0.4568	1.971	99.38
6	5600	5.188e-06	5.176e-06	0.09773	0.4572	2.001	99.37
7	1600	5.22e-06	5.209e-06	0.1144	0.4537	2.046	99.37
8	200	4.982e-06	4.954e-06	0.2548	0.444	2.122	99.33



Remaining error by device case

Same attribution as in section 7 (one device case replaced by exact single-FET SPICE currents at the gate bias points):

exact_case	median_err_pct	p95_err_pct
nmos_on	0.4406	2.373
nmos_off	0.42	2.122
pmos_on	0.433	2.434
pmos_off	0.3128	1.743

With ML everywhere: gate median error 0.53 %.

7. Error analysis

The analysis below is for the specified 10,000-sample run (sections 2-5); the dataset-size study covers all runs.

Which device model limits the gate accuracy

Every transistor bias point that occurs inside the 18 gates x 200 corners was re-simulated as a single FET in SPICE. Replacing the ML currents of one device case by the exact SPICE currents (all other cases stay ML) gives the remaining gate-level error:

exact_case	median_err_pct	p95_err_pct
nmos_on	1.003	5.198
nmos_off	0.8841	5.045
pmos_on	0.9686	5.301
pmos_off	0.4941	2.118

With ML everywhere the same computation gives median 1.09 %. The case whose replacement removes the most error is the one that limits the gate accuracy. Inside gates, Vd and Vs sit at the rails (0 or VDD), i.e. at the corners of the uniform Vd-Vs training square, where the training data is thinnest; device errors at these bias points are larger than on the random test set.

Why pmos_off is the weakest device model

Inside the gates every OFF PMOS has Vs = Vg = Vb = VDD, so only (Vd - Vs) Id enters the leakage; that corner (Vs > 0.95 VDD) holds about 5 % of the uniform training data (`experiments/exp8_pmos_off_diag.py`). Variants trained on the same 10k data and split, scored on validation and on the independent rail-bias set (`corner` = Vs > 0.95 VDD; the gate test corners are not used); errors in %:

variant	train_med	val_med	val_power_med	val_corner_power_med	rail_corner_power_med
base (exp9)	0.149	1.045	0.767	1.981	3.375
wide (exp9)	0.006	1.222	0.917	2.230	3.615
deep (exp9)	0.070	1.135	0.893	1.494	2.420
long (exp9)	0.037	0.903	0.676	1.988	3.111
wd (exp12)	0.152	1.032	0.780	2.413	3.466
ens15 (exp12)	0.095	0.758	0.524	1.794	2.852
small (exp12)	0.758	1.181	0.882	2.111	3.827
tunnel (exp12)	0.132	1.089	0.824	1.793	4.191

- The training error is 7-193x below the validation error (except for the deliberately smaller network): the model fits its training set far better than new data, i.e. it is variance (data) limited.
- Against the base network (validation median 1.045 %, rail-corner power median 3.375 %), lower validation error: ens15 (exp12) 0.758 %, long (exp9) 0.903 %, wd (exp12) 1.032 %; lower rail-corner error: deep (exp9) 2.420 %, ens15 (exp12) 2.852 %, long (exp9) 3.111 %. The best variant on validation (ens15 (exp12)) lowers the validation error by 27 % and the rail-corner error by 15 %; doubling the data to 20,000 samples per case with the base settings lowers the validation error by 45 % (learning curve below).
- The SPICE data itself is smooth: re-simulating a training row reproduces it to 2e-16 relative, and narrow one-parameter sweeps deviate from a smooth polynomial by at most 0.0039 % in device power (`experiments/exp10_spice_smoothness.py`).
- Learning BSIM4's gate-overlap tunnelling currents (Igd, Igs) as separate targets and rebuilding the terminal currents was worse for every case (validation median, terminal vs separated targets: nmos_on 0.23 / 0.26 %, nmos_off 0.23 / 0.30 %, pmos_on 0.18 / 0.24 %, pmos_off 1.04 / 1.92 %): in OFF devices Igd is practically the whole drain current (median |Igd / Id| = 1.0000 in nmos_off, median |Igd / Id| = 1.0000 in pmos_off), so the split adds nothing (`experiments/exp11_component_targets.py`).
- The ens15 (exp12) variant costs 3.8x the training time of the base.

Full analysis output

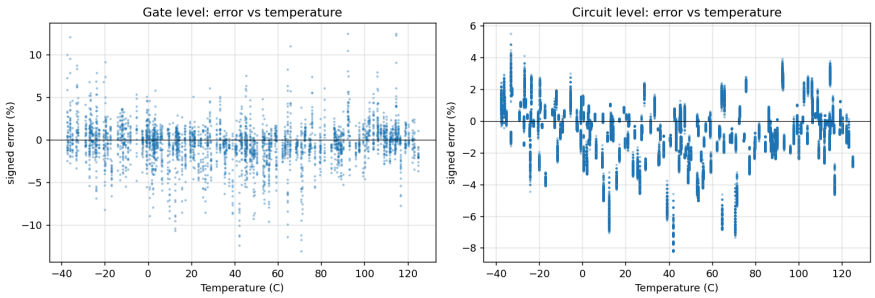
```
== Device level (test sets): relative error % by region of |Vd-Vs|/VDD (pooled over id,ig,is,ib)
nmos_on   |Vds|/VDD in [0.00,0.05): n= 144 median    1.21%  p95   192.59%  (Id,Is only: median 2.92%, p95 830.37%)
nmos_on   |Vds|/VDD in [0.05,0.30): n= 609 median    0.28%  p95     3.84%  (Id,Is only: median 0.31%, p95 4.31%)
nmos_on   |Vds|/VDD in [0.30,1.01): n= 747 median    0.17%  p95     0.80%  (Id,Is only: median 0.19%, p95 0.70%)
nmos_on   Spearman(median err over id,ig,is ; temp) = -0.051
nmos_on   Spearman(median err over id,ig,is ; |vth0|) = -0.004
nmos_off  |Vds|/VDD in [0.00,0.05): n= 158 median    0.27%  p95    13.92%  (Id,Is only: median 0.47%, p95 64.47%)
nmos_off  |Vds|/VDD in [0.05,0.30): n= 603 median    0.20%  p95     1.66%  (Id,Is only: median 0.33%, p95 2.21%)
nmos_off  |Vds|/VDD in [0.30,1.01): n= 739 median    0.27%  p95     2.74%  (Id,Is only: median 0.47%, p95 4.15%)
nmos_off  Spearman(median err over id,ig,is ; temp) = +0.132
nmos_off  Spearman(median err over id,ig,is ; |vth0|) = -0.048
pmos_on   |Vds|/VDD in [0.00,0.05): n= 148 median    0.82%  p95    88.84%  (Id,Is only: median 1.18%, p95 169.77%)
pmos_on   |Vds|/VDD in [0.05,0.30): n= 623 median    0.24%  p95     4.10%  (Id,Is only: median 0.25%, p95 2.26%)
pmos_on   |Vds|/VDD in [0.30,1.01): n= 729 median    0.14%  p95     0.64%  (Id,Is only: median 0.14%, p95 0.57%)
pmos_on   Spearman(median err over id,ig,is ; temp) = -0.081
pmos_on   Spearman(median err over id,ig,is ; |vth0|) = -0.008
pmos_off  |Vds|/VDD in [0.00,0.05): n= 144 median    1.28%  p95    37.51%  (Id,Is only: median 1.80%, p95 76.30%)
pmos_off  |Vds|/VDD in [0.05,0.30): n= 612 median    0.95%  p95     6.50%  (Id,Is only: median 1.34%, p95 9.36%)
pmos_off  |Vds|/VDD in [0.30,1.01): n= 744 median    1.13%  p95     9.11%  (Id,Is only: median 1.83%, p95 14.36%)
pmos_off  Spearman(median err over id,ig,is ; temp) = -0.242
pmos_off  Spearman(median err over id,ig,is ; |vth0|) = +0.020

== Gate level: Spearman(|err|, parameter)
  temp       rho = -0.045   (signed err: -0.026)
  vdd        rho = +0.175   (signed err: +0.079)
  vth0_n     rho = -0.009   (signed err: -0.084)
  vth0_p_abs rho = -0.071   (signed err: +0.086)
  toxe_n     rho = +0.148   (signed err: -0.174)
  toxe_p     rho = +0.065   (signed err: -0.227)
  u0_n       rho = -0.014   (signed err: -0.010)
  u0_p       rho = -0.016   (signed err: +0.078)
  ndep_n     rho = +0.018   (signed err: -0.001)
  ndep_p     rho = -0.052   (signed err: +0.017)
  xj_n       rho = -0.005   (signed err: +0.025)
  xj_p       rho = -0.008   (signed err: +0.064)
== Gate level: |err| % by temperature quintile
                  median    p95
tbin
(-37.57, -1.236]    1.202  5.016
(-1.236, 22.555]    0.942  5.057
(22.555, 54.076]    1.222  5.808
(54.076, 90.532]    1.102  5.596
(90.532, 124.827]   0.973  4.200
== Gate level: worst 8 rows
      gate pattern  corner        temp       vdd         spice            ml        err
1571  NOR2      01     171   70.704368  0.818614  2.319354e-07  2.016213e-07 -13.070063
1501  NOR2      01     101  114.474119  0.974839  4.457799e-07  5.012883e-07  12.451972
1818  NOR2      11      18   92.393490  0.988256  9.740689e-08  1.095070e-07  12.422255
204    INV       1       4   42.034284  0.986133  6.670521e-08  5.843664e-08 -12.395702
1901  NOR2      11     101  114.474119  0.974839  9.933387e-08  1.115027e-07  12.250446
1734  NOR2      10     134  -36.187723  0.846844  6.480399e-08  7.261544e-08  12.053959
1771  NOR2      10     171   70.704368  0.818614  1.578956e-07  1.394072e-07 -11.709256
341    INV       1     141   64.442890  0.947318  7.507192e-08  6.651494e-08 -11.398374

== Circuit level: Spearman(|err|, parameter)
  temp       rho = +0.053   (signed err: -0.201)
  vdd        rho = +0.178   (signed err: +0.043)
  vth0_n     rho = +0.005   (signed err: -0.108)
  vth0_p_abs rho = -0.149   (signed err: +0.106)
  toxe_n     rho = +0.233   (signed err: -0.313)
  toxe_p     rho = +0.221   (signed err: -0.386)
  u0_n       rho = +0.003   (signed err: -0.005)
  u0_p       rho = -0.010   (signed err: +0.076)
  ndep_n     rho = -0.042   (signed err: -0.007)
  ndep_p     rho = -0.030   (signed err: +0.068)
  xj_n       rho = +0.002   (signed err: +0.051)
  xj_p       rho = -0.058   (signed err: +0.095)
== Circuit level: |err| % by temperature quintile
                  median    p95
tbin
(-37.57, -1.236]    0.906  2.875
(-1.236, 22.555]    0.875  4.043
(22.555, 54.076]    1.526  4.967
(54.076, 90.532]    1.386  5.130
(90.532, 124.827]   0.838  2.939
== Circuit level: worst 8 rows
       vec  corner       temp       vdd     spice        ml       err
34804  174       4  42.034284  0.986133  0.000003  0.000002 -8.202014
22004  110       4  42.034284  0.986133  0.000003  0.000003 -8.193822
34604  173       4  42.034284  0.986133  0.000003  0.000003 -8.193822
31604  158       4  42.034284  0.986133  0.000003  0.000003 -8.193821
34204  171       4  42.034284  0.986133  0.000003  0.000002 -8.191647
46804  234       4  42.034284  0.986133  0.000003  0.000002 -8.191647
21804  109       4  42.034284  0.986133  0.000003  0.000003 -8.186321
18804   94       4  42.034284  0.986133  0.000003  0.000003 -8.186321

== Device models at the bias points used inside the gates (SPICE single-FET reruns)
nmos_off  n= 4200  power-term error: median 1.335%  p95 5.35%  bias -0.195%   | Ig error median 0.701%
nmos_on   n= 4200  power-term error: median 1.110%  p95 7.14%  bias +0.020%   | Ig error median 1.050%
pmos_off  n= 4200  power-term error: median 3.203%  p95 14.19%  bias -0.838%   | Ig error median 2.008%
pmos_on   n= 4200  power-term error: median 0.971%  p95 3.89%  bias +0.192%   | Ig error median 1.004%

== Attribution (gate level): ML everywhere -> median |err| 1.089%, p95 5.24%
   with exact SPICE currents for nmos_on  : median |err| 1.003%, p95 5.20%
   with exact SPICE currents for nmos_off : median |err| 0.884%, p95 5.04%
   with exact SPICE currents for pmos_on  : median |err| 0.969%, p95 5.30%
   with exact SPICE currents for pmos_off : median |err| 0.494%, p95 2.12%
```



Dataset size (learning curve)

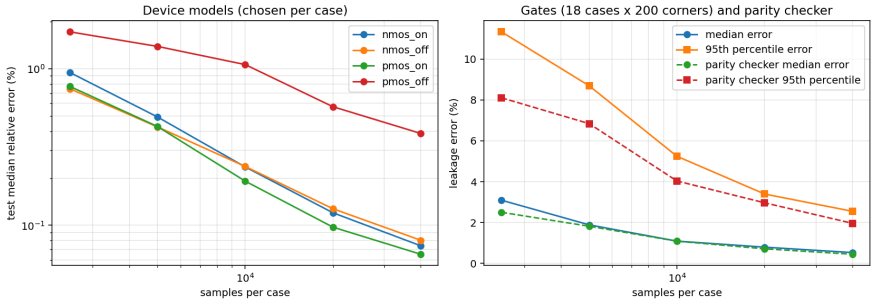
Each row is an independent full run (new data, retraining, the same 200 gate corners). The parity-checker columns are the hierarchical sum of that run's gate estimates against the same full-circuit SPICE reference (the SPICE side does not depend on the models):

samples_per_case	gate_mean_accuracy_pct	gate_median_err_pct	gate_p95_err_pct	circuit_mean_accuracy_pct	circuit_median_err_pct	nmos_on_test_median_pc
2500	95.892	3.093	11.337	96.781	2.500	0.94
5000	97.175	1.885	8.694	97.486	1.816	0.49
10000	98.355	1.089	5.244	98.589	1.086	0.23
20000	98.850	0.793	3.394	99.008	0.712	0.12
40000	99.191	0.528	2.550	99.380	0.447	0.07

Factor by which the device test median error changes per doubling of the data: nmos_on 0.48-0.62; nmos_off 0.54-0.63; pmos_on 0.45-0.67; pmos_off 0.54-0.81. Every device error falls with every doubling and the gate and circuit errors follow, so the remaining error is mainly a data (variance) limit, not a model-form limit. Of the sizes tested (2,500, 5,000, 10,000, 20,000, 40,000 samples per case), the smallest that reaches 99 % mean accuracy is 40,000 at gate level and 20,000 for the parity checker.

Best device validation median error (%) per size, the basis for the size of the final models (section 6):

samples_per_case	nmos_on_val_median_pct	nmos_off_val_median_pct	pmos_on_val_median_pct	pmos_off_val_median_pct
2500	1.024	0.801	0.762	2.031
5000	0.492	0.426	0.437	1.341
10000	0.232	0.232	0.182	1.045
20000	0.123	0.130	0.099	0.571
40000	0.076	0.081	0.065	0.382



8. Limitations and improvements

- The device models are trained on Vg fixed at exactly 0 or VDD; inside AND2/OR2 the output INV sees a near-rail (not exact) input voltage, a small extrapolation.
- Relative device error is largest where the true current is tiny or crosses zero (|Vds| near 0); such points rarely carry gate leakage, but they dominate the device-level 95th percentile.
- The device bias points inside gates lie at the rails, where uniform Vd/Vs sampling places few samples; sampling more densely near the rails (in addition to the uniform data, which the specification requires) would target the gate-level error directly.
- At the specified 10,000 samples per case the pmos_off model is variance limited; the dataset-size study shows the 99 % level is reached with more samples per case, at a proportionally longer data-generation and training time.
- The hierarchical circuit estimate ignores gate-to-gate loading by construction; the SPICE gate-sum row in section 5 measures that part separately.
- The pmos_off model has the largest pooled test error (1.06 % median); making the pmos_off currents exact lowers the gate median error from 1.09 % to 0.49 %, the largest single gain.